

HR Attrition Analysis

A data-driven investigation into employee attrition patterns to support strategic retention decisions.

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Tool	MySQL (data cleaning + analysis)
Dataset	Simulated HR dataset — 750 employees, 8 departments
Audience	Chief Human Resources Officer (CHRO)
Purpose	Identify attrition risk areas and provide actionable retention recommendations

750

Employees Analysed

8

Departments

115

Attrition Records

5

Business Questions

01 Business Context

The Chief Human Resources Officer (CHRO) has raised concerns about employee attrition heading into year-end. She needs to understand **who is leaving, from where, and why** — so that a limited retention budget can be directed at the right departments and root causes rather than applied uniformly across the organisation.

This analysis works through five targeted business questions, moving from broad department-level attrition rates down to salary comparisons between leavers and stayers at the job-role level. Every finding is grounded in the data — interpretations are clearly flagged as such.

02 Dataset & Data Quality

The dataset comprises four related tables: **employees**, **departments**, **salary_history**, and **attrition**. The raw data contained the following quality issues, identified during the audit phase:

Issue	Detail	Resolution
Inconsistent job_role casing	Same roles stored as 'HR Manager', 'HR MANAGER', 'hr manager'	LOWER() applied in clean view
Orphaned department_id	~2% of rows pointed to dept_id = 99 (non-existent)	Inferred correct dept from 1-to-1 job_role mapping
NULL department_id	~3% of employees had no department recorded	Same inference approach as orphaned FK rows
Duplicate employee rows	Same person entered twice with different employee_id	ROW_NUMBER() deduplication, kept earliest ID
NULL performance_rating	~4% of employees unrated	Kept in view; excluded only from Q3 (performance analysis)
Bad salary values	60 rows with negative, zero, or NULL salary_amount	Excluded from salary_clean view
Fully NULL garbage row	One row with every column NULL including employee_id	Excluded via WHERE employee_id IS NOT NULL
Missing attrition records	3 terminated employees had no row in attrition table	Accepted as irrecoverable data gap; noted in analysis

Cleaning approach: Two SQL views were created — *employees_clean* and *salary_clean* — applying all fixes at query time without modifying the source tables. Raw data remains intact and fully reproducible.

03 Analysis & Findings

Q1 — Which departments have the highest attrition rate?

Attrition rate = (employees who left ÷ total department headcount) × 100. A LEFT JOIN to the attrition table ensures active employees remain in the denominator.

Department	Total Employees	Leavers	Attrition Rate
Sales	85	18	21.18%
Human Resources	98	17	17.35%
Operations	86	14	16.28%
Engineering	88	14	15.91%
Legal	101	16	15.84%
Finance	97	13	13.40%
Customer Support	98	13	13.27%
Marketing	97	9	9.28%

Finding: Sales has the highest attrition rate at 21.2% — approximately 1 in 5 employees left during the analysis period. Marketing has the lowest at 9.3%, less than half the Sales rate. Department-level rates vary significantly, confirming that company-wide averages mask very different retention realities across teams.

Q2 — Is attrition voluntary or involuntary, and does the split vary?

Voluntary = employee chose to leave. Involuntary = company decision (fired, restructured). A CTE was used to fix the department headcount denominator before splitting by exit type — without this, the GROUP BY exit_type would scope the denominator incorrectly and produce 100% rates.

Department	Exit Type	Leavers	Rate
Human Resources	Voluntary	14	14.29%
Sales	Voluntary	11	12.94%
Engineering	Voluntary	11	12.50%
Legal	Voluntary	11	10.89%
Operations	Voluntary	9	10.47%
Sales	Involuntary	7	8.24%
Finance	Voluntary	9	9.28%
Human Resources	Involuntary	3	3.06%

Finding: Human Resources has the highest voluntary attrition (14.3%) — employees are actively choosing to leave, which points to internal dissatisfaction rather than company-driven exits. Sales is uniquely problematic: it has high rates on BOTH sides (12.9% voluntary + 8.2% involuntary), suggesting a department under pressure from multiple directions simultaneously.

Q3 — Are high performers leaving at a higher rate than low performers?

Employees grouped by performance rating (1 = lowest, 5 = highest). NULL ratings excluded as they cannot be meaningfully placed in a performance-based comparison.

Performance Rating	Total Employees	Leavers	Attrition Rate
5 — Exceptional	97	19	19.59%
4 — Strong	215	33	15.35%
2 — Below Avg	74	11	14.86%
3 — Average	297	43	14.48%
1 — Poor	37	5	13.51%

Critical finding: Top performers (rating 5) have the HIGHEST attrition rate at 19.6%, while the lowest performers are the most stable at 13.5%. The company is retaining its weakest employees while losing its best — a pattern that degrades the talent base over time. This requires immediate investigation.

Q4 — What is the salary gap between leavers and stayers per job role?

Most recent salary per employee (via ROW_NUMBER on effective date) used to avoid distorting averages with historical raises. CASE WHEN inside AVG() pivots leavers and stayers into separate columns. Positive gap = stayers earn more. Negative gap = leavers earned more.

Job Role	Avg Salary (Stayers)	Avg Salary (Leavers)	Gap
Compliance Officer	705,504	867,500	-161,996
Operations Manager	861,844	1,023,567	-161,723
DevOps Engineer	852,979	990,567	-137,588
Sales Manager	954,396	1,054,933	-100,537
HR Manager	842,941	741,100	+101,841
Legal Counsel	1,088,550	979,625	+108,925
Marketing Manager	873,638	759,580	+114,058
Account Manager	623,247	506,900	+116,347

Finding: Most roles show a negative salary gap — leavers were earning MORE than the stayers who replaced them. This rules out pay as the primary attrition driver for high-exit roles like Sales and Operations. Root causes are likely non-monetary: management quality, career growth ceiling, or burnout. Roles with a positive gap (Account Manager, Marketing Manager) suggest those leavers may have felt underpaid relative to peers.

Q5 — Which departments have the longest average tenure before employees leave?

Tenure calculated as DATEDIFF(exit_date, hire_date) / 365. Only employees with a confirmed exit record are included. Shorter tenure = employees deciding to leave earlier.

Department	Avg Tenure Before Leaving
Finance	2.37 years
Legal	2.66 years
Sales	2.80 years
Customer Support	3.13 years
Marketing	3.32 years
Human Resources	3.42 years
Operations	3.44 years
Engineering	3.91 years

Finding: Finance and Legal employees leave after only 2.4–2.7 years on average — the shortest tenure in the organisation. Combined with their salary gap data, this suggests early disillusionment rather than a long-simmering pay dispute. Engineering retains leavers the longest (3.9 years), suggesting stronger engagement or more complex career decisions in technical roles.

04 Recommendations

Based on the five findings above, the following actions are recommended in order of urgency:

01 Investigate Sales Management Structure — Urgent

Sales has the highest overall attrition (21%), high voluntary AND involuntary rates simultaneously, and leavers were earning more than stayers — ruling out pay as the fix. The CHRO should conduct structured exit interviews and review how sales targets are set, managed, and enforced. This is a leadership and culture problem, not a compensation one.

02 Address Top-Performer Flight Risk — Urgent

Rating-5 employees are leaving at 19.6% — the highest attrition rate of any group. These are the employees most likely to be poached and hardest to replace. The CHRO should audit promotion pathways, recognition programmes, and workload distribution for high performers across all departments before more institutional knowledge walks out.

03 Investigate Human Resources Department

HR has the highest voluntary attrition (14.3%) — a department that exists to retain others is itself struggling to retain its own people. Short-term: conduct anonymous pulse surveys within the HR team. Medium-term: review team structure and management quality.

04 Act Early in Finance and Legal

Employees in Finance and Legal are leaving after only 2.4–2.7 years. The company is losing people before they reach full productivity. Consider structured 18-month check-ins, clearer career progression frameworks, and mentorship programmes targeted at junior-to-mid level staff in these departments.

05 Maintain Marketing Approach

Marketing has the lowest attrition rate (9.3%) and reasonable tenure figures. Before assuming this is a pay issue, the CHRO should document what Marketing is doing differently — management style, team culture, target-setting — and use it as a benchmark for struggling departments.

05 Methodology & Limitations

Tools used:

MySQL (MySQL Workbench) for all data cleaning, view creation, and analysis queries. ReportLab (Python) for this case study document.

Analytical approach:

All analysis was performed on two clean views (employees_clean, salary_clean) built on top of the raw tables. No raw data was modified. Cleaning decisions were documented and justified at each step. Business questions were framed before queries were written.

Limitations:

- This is a simulated dataset. Findings are illustrative of real analytical patterns but do not represent an actual organisation.
- 3 terminated employees had no attrition record — their exit type and reason are unknown and they are excluded from Q2.
- ~4% of employees have no performance rating. These are excluded from Q3 only; they remain in all other analyses.
- Salary gap analysis uses the most recent recorded salary, which may not reflect total compensation (bonuses, equity, benefits).
- Attrition rate calculations use a static snapshot (total headcount) rather than a rolling monthly average — a limitation of the available data granularity.